

FORMULATION AND ESTIMATION OF DYNAMIC MODELS USING PANEL DATA

T.W. ANDERSON*

Stanford University, Stanford, CA 94305, USA

Cheng HSIAO**

Bell Laboratories, Murray Hill, NJ 07974, USA

University of Toronto, Toronto, Ont., Canada M5S 1A1

This paper presents a statistical analysis of time series regression models for longitudinal data with and without lagged dependent variables under a variety of assumptions about the initial conditions of the processes being analyzed. The analysis demonstrates how the asymptotic properties of estimators of longitudinal models are critically dependent on the manner in which samples become large: by expanding the number of observations per person, holding the number of people fixed, or by expanding the number of persons, holding the number of observations per person fixed. The paper demonstrates which parameters can and cannot be identified from data produced by different sampling plans.

1. Introduction

A common practice in econometrics is to assume that the large number of factors which affect the dependent variables, but are not explicitly included as explanatory variables, may be suitably summarized by random disturbances. When a number of individuals are observed over several periods of time, it seems appropriate to decompose the error term into two components; factors affect specific individual units in more or less the same fashion over time and factors affect both individual units and time periods. This may lead to an error component formulation in the econometric analysis of cross-sectional and time series data [e.g., Balestra and Nerlove (1966), Mundlak (1961, 1978)].

Various aspects of several formulations of error components models have

*Research supported by National Science Foundation Grant SES79-13976 at the Institute for Mathematical Studies in the Social Sciences, Stanford University.

**Research supported by National Science Foundation Grant SES80-07576 at the Institute for Mathematical Studies in the Social Sciences, Stanford University, and by Social Sciences and Humanities Research Council of Canada Grant 410-80-0080 at the Institute for Policy Analysis, University of Toronto.

been studied, for example, by Amemiya (1967), Hausman and Taylor (1980), Lee (1979), Maddala (1971b), Mundlak (1978), Nerlove (1971), Swamy and Arora (1972), Taylor (1980), Trognon (1978), Wallace and Hussain (1969). In this paper we present a further analysis of the dynamic error components models of the form¹

$$y_{it} = \beta y_{i,t-1} + \underset{1 \times K_1}{\delta'} \underset{K_1 \times 1}{z_i} + \underset{1 \times K_2}{\gamma'} \underset{K_2 \times 1}{x_{it}} + v_{it}, \quad \begin{matrix} i = 1, \dots, N, \\ t = 1, \dots, T, \end{matrix} \quad (1.1)$$

where $v_{it} = \alpha_i + u_{it}$, $E\alpha_i = Eu_{it} = 0$, $E\alpha_i z_i' = 0'$, $E\alpha_i x_{it}' = 0'$, $E\alpha_i u_{jt} = 0$, $E\alpha_i \alpha_j = \sigma^2$ for $i=j$, and $=0$ for $i \neq j$, $Eu_{it} u_{js} = \sigma_u^2 = \lambda \sigma^2$ for $i=j$, $t=s$, and $=0$ otherwise, z_i is a $K_1 \times 1$ vector of time-invariant exogenous variables, x_{it} is a $K_2 \times 1$ vector of time-varying exogenous variables, β is 1×1 , and δ and γ are $K_1 \times 1$ and $K_2 \times 1$ vectors of parameters, respectively. We also assume that $|\beta| < 1$ and α_i and u_{it} are normally distributed (although these assumptions are not always necessary). Close attention is paid here to the properties of various estimators in the way in which the number of time series observations T or the cross-sectional units N tend to infinity. Contrary to the case of the dynamic model for a single time series, the assumption about the initial observations plays a crucial role in interpreting the model and devising consistent estimates. In an earlier paper [Anderson and Hsiao (1981a)] we have focused on (i) the interpretation and (ii) the asymptotic properties of the estimators of simple time series models arising from different assumptions about the initial observations. In this paper we generalize the analysis to include exogenous variables. Such generalization provides interesting and sometimes quite different perspectives from pure time series models.

2. Formulation of models with time-invariant exogenous variables

In many economic models time-invariant exogenous variables such as sex, race, schooling, marital status, and socioeconomic background variables are present. In this section we consider the formulation of dynamic models containing time-invariant exogenous variables only. We assume

$$y_{it} = \beta y_{i,t-1} + \delta' z_i + \alpha_i + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (2.1)$$

where δ and z_i are $K_1 \times 1$ vectors of parameters and time-invariant exogenous variables.²

¹Mundlak (1978) has suggested a formulation which allows correlation between α_i and exogenous variables. His formulation leads to the fixed effect estimation of (1.1). The maximum likelihood estimator under the assumption that α_i are fixed constants is the covariance estimator under our assumption that α_i are random and uncorrelated with exogenous variables. Since the properties of the covariance estimator will be dealt with in the paper, for simplicity we shall ignore an explicit formulation of Mundlak's (1978) model in this paper.

²In the special case where $K_1 = 1$ and $z_i = 1$ we have a time series model with mean (intercept) not assumed zero.

Just as in the case of time series models analyzed earlier [Anderson and Hsiao (1981a)], we may have four different types of models associated with (2.1) under different assumptions about the initial observations. The first type of model is a conventional one where we assume that y_{i0} are observed fixed constants (Case I). Rewrite (2.1) in the form

$$(y_{it} - \tilde{\delta}' z_i - \eta_i) = \beta(y_{i,t-1} - \tilde{\delta}' z_i - \eta_i) + u_{it}, \quad (2.2)$$

where

$$\alpha_i = (1 - \beta)\eta_i, \quad E\eta_i = 0, \quad \text{var}(\eta_i) = \sigma_\eta^2 = \sigma^2/(1 - \beta)^2, \quad \delta = \tilde{\delta}(1 - \beta).$$

We can see that this assumption implies that a cross-sectional unit may start at some arbitrary y_{i0} and gradually move towards a level of $(\alpha_i + \tilde{\delta}' z_i)/(1 - \beta) = \tilde{\delta}' z_i + \eta_i$ according to a probability law. The individual level is determined by the observed characteristics z_i and the unobservable effect η_i . The unobservable individual effect is assumed to be a random draw from a population with mean zero and variance σ_η^2 . This appears to be a reasonable model, although there might be a question of treating y_{i0} as fixed if the decision of when to start sampling is arbitrary, in particular, independent of the value of y_{i0} , because it implies $E\alpha_i y_{i0} = 0$. The unobservable individual effects are not brought into the model at time 0, but affect the process at time 1 and later.

An alternative formulation is to assume y_{i0} random with the mean determined by the observable characteristics z_i . For instance, we may let $y_{i0} = \tilde{\delta}' z_i + \varepsilon_i$, $i = 1, \dots, N$, where the random variable ε_i may be viewed as the effect of (unobserved) initial individual endowments. Depending on the assumption about ε_i , the impact of initial endowments will change as time goes by. If we assume ε_i random with mean zero and variance σ_ε^2 and independent of α_i and u_{it} (Case IIa), its impact gradually diminishes and eventually vanishes. The model is somewhat like the first model in which the starting value corrected for the mean $\tilde{\delta}' z_i$ is independent of the equilibrium level η_i , except that now the starting observable value is not a fixed constant but a random draw from a population with mean $\tilde{\delta}' z_i$ and variance σ_ε^2 . On the other hand, if we wish to allow the initial endowment to affect the equilibrium level (Case IIb), we may let $y_{i0} = \tilde{\delta}' z_i + \alpha_i$, $i = 1, \dots, N$. Then, as time goes by, the effect of initial endowments cumulates and eventually reaches a level of $\alpha_i/(1 - \beta) = \eta_i$.

A different formulation allowing dependence between α_i and y_{i0} is to rewrite (2.1) as

$$(w_{it} - \tilde{\delta}' z_i) = \beta(w_{i,t-1} - \tilde{\delta}' z_i) + u_{it}, \quad t = 1, \dots, T, \quad i = 1, \dots, N, \quad (2.3)$$

$$y_{it} = w_{it} + \eta_i, \quad t = 0, 1, \dots, T, \quad i = 1, \dots, N, \quad (2.4)$$

or

$$w_{it} = \beta w_{i,t-1} + u_{it}, \quad t = 1, \dots, T, \quad i = 1, \dots, N, \quad (2.5)$$

$$y_{it} = w_{it} + \delta' z_i + \eta_{it}, \quad t = 0, 1, \dots, T, \quad i = 1, \dots, N, \quad (2.6)$$

where w_{it} , η_{it} and u_{it} are unobservable. Although (2.3) implies that the unobservable process $\{w_{it}\}$ has mean $\delta' z_i$ and is independent of η_{it} , and (2.5) implies that the level $\delta' z_i + \eta_{it}$ and the unobservable process $\{w_{it}\}$ are independent, in practice there is no way to distinguish these two cases based on observed values of y and z only. We shall, therefore, make no attempt to distinguish them. Later (section 8) when we introduce time-varying exogenous variables into the model we shall separate these two cases in terms of state dependence and serial correlation.

Analogous to the assumption that y_{i0} may be fixed or random, we may assume that w_{i0} are either fixed or random. When w_{i0} are assumed fixed (Case III), the model is similar to the first one where an individual may start at some value y_{i0} and move towards a level of $\delta' z_i + \eta_{it}$, except that in this case the individual effect η_i affects y_{i0} .

When w_{i0} are assumed random (Case IV), alternative standard assumptions are (a) stationary with variance $\sigma^2/(1-\beta^2)$ or (b) random with arbitrary variance. In sub-case (a) y_{i0} is considered to have the marginal normal distribution determined by the stationary distribution; that is, y_{i0} is viewed to form part of a stationary process as any other y_{it} . In sub-case (b) the starting value y_{i0} is random from a population which may not have the same marginal distribution as at later periods. Since, as demonstrated by Anderson and Hsiao (1981a), sub-case (b) does not lead to simpler manipulation, we shall not follow the assumption that w_{i0} is random with arbitrary variance in this paper.

3. Identification conditions for models with time-invariant exogenous variables

Identification is a fundamental problem concomitant with the existence of a structure. It is a property of the parameterization of the distribution of observations. Because of the nature of the problem considered here, the identification of a model may depend on the numbers of time series observations T and of the cross-sectional units N .

Using theorems derived by Rothenberg (1971) and Richmond (1974), we can show the following:³

Proposition 3.1. In Cases I and IIa, λ and σ^2 are not identifiable if $T=1$.

In Cases I and IIa when $T=1$ we have N independently distributed random variables $v_{i1} = \alpha_i + u_{i1}$ with variance $\sigma^2(1+\lambda)$. It is not possible to distinguish λ and σ^2 . However, it is possible to identify λ and σ^2 for Cases

³For the detail of the derivation, see Anderson and Hsiao (1981b).

IIb, III and IVa because under these assumptions y_{i0} and y_{i1} contain separate information on λ and σ^2 (or σ_u^2 and σ_η^2). For ease of exposition in this paper we shall assume that $T \geq 2$ for Cases I and IIa, and $T \geq 1$ for Cases IIb, III, and IVa.

Proposition 3.2. A necessary and sufficient condition for the identification of dynamic error components models with time-invariant exogenous variables is that $\sum_{i=1}^N \mathbf{z}_i \mathbf{z}_i'$ has rank K_1 .

From Proposition 3.2 we have the following necessary condition for the identification of the model:

Proposition 3.3. A necessary condition for the identification of dynamic error components model with K_1 time-invariant exogenous variables is $N \geq K_1$.

It is interesting to note that whether the complete model is identifiable or not, the parameter β is always identifiable.

Proposition 3.4. β is identifiable.

4. Estimation of models with time-invariant exogenous variables

In this section we assume that the model is identified and discuss the estimation methods. Two methods are widely used in the estimation of error components models: the covariance method and the maximum likelihood method. For the dynamic models considered in section 2, the consistency of these two methods depends critically on the assumption about the initial conditions and on the way in which the number of time series observations T or the cross-sectional units N tends to infinity. We now examine the consistency property of various estimation methods under various assumptions.

Rewrite (2.1) in vector form as

$$\mathbf{y}_i = \mathbf{y}_{i,-1} \beta + \delta' \mathbf{z}_i \mathbf{e} + \mathbf{v}_i, \quad i = 1, \dots, N, \quad (4.1)$$

where $\mathbf{y}_i = (y_{i1}, \dots, y_{iT})'$, $\mathbf{y}_{i,-1} = (y_{i0}, \dots, y_{i,T-1})'$, $\mathbf{v}_i = (v_{i1}, \dots, v_{iT})'$, and $\mathbf{e}$ denotes the $T \times 1$ vector $(1, \dots, 1)'$. The idea of the covariance estimator (CV) is to eliminate the time-invariant individual effect α_i by premultiplying (4.1) by $\mathbf{Q} = \mathbf{I}_T - (1/T)\mathbf{e}\mathbf{e}'$. However, the multiplication of (4.1) by $\mathbf{Q}$ also eliminates all time-invariant variables $\mathbf{z}_i$. Thus, their coefficients δ are not estimable. The CV can estimate only β by solving the following normal equation:

$$\sum_{i=1}^N \left[\frac{1}{T} \sum_{t=1}^T y_{it,t-1}^2 - \bar{y}_{i,-1}^2 \right] \hat{\beta}_{CV} = \sum_{i=1}^N \left[\frac{1}{T} \sum_{t=1}^T y_{it} y_{i,t-1} - \bar{y}_i \bar{y}_{i,-1} \right], \quad (4.2)$$

where $\bar{y}_i = \sum_{t=1}^T y_{it}/T$, $\bar{y}_{i,-1} = \sum_{t=1}^T y_{i,t-1}/T$. As shown by Anderson and Hsiao (1981a), (4.2) is equivalent to applying the least squares estimation of β to the model $y_{i,t} = \beta y_{i,t-1} + d_i + u_{it}$, where d_i are treated as unknown constants. It is consistent when T tends to infinity (regardless of whether N is fixed or tends to infinity) and inconsistent when T is fixed but N tends to infinity. The CV of β is also the MLE under the assumption that y_{i0} and α_i are fixed constants and u_{it} are normally distributed.

When both α_i and u_{it} are random and normally distributed, the log likelihood function under the assumption that y_{i0} are fixed [eq. (2.2)] is

$$\begin{aligned} \log L = & -\frac{NT}{2} \log 2\pi - \frac{NT}{2} \log \sigma^2 - \frac{N}{2} \log |A| \\ & - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - y_{i,-1} \beta - \delta' z_i e)' A^{-1} (y_i - y_{i,-1} \beta - \delta' z_i e), \end{aligned} \quad (4.3)$$

where

$$\begin{aligned} A &= \lambda I_T + ee', \quad |A| = \lambda^{T-1}(T + \lambda), \\ A^{-1} &= \frac{1}{\lambda} I_T - \frac{1}{\lambda(T + \lambda)} ee' = \frac{1}{\lambda} Q + \frac{1}{T(T + \lambda)} ee'. \end{aligned} \quad (4.4)$$

The maximum likelihood estimator (MLE) is obtained by taking partial derivatives of (4.3) with respect to β , δ' , σ^2 and λ . When N is fixed and T tends to infinity, as shown by Amemiya (1967), $\hat{\lambda}_{MLE}/(\hat{\lambda}_{MLE} + T) \rightarrow 0$ as $T \rightarrow \infty$; therefore $\hat{\beta}_{MLE} - \hat{\beta}_{CV}$ converges stochastically to 0 and $\hat{\beta}_{MLE}$ is consistent.⁴ Using the fact that $\text{plim}_{T \rightarrow \infty} \hat{\beta} = \beta$, we solve for $\hat{\delta}_{MLE}$ and find

$$\begin{aligned} \hat{\delta}_{MLE} = & \left\{ \delta + \left[\sum_{i=1}^N z_i z_i' \right]^{-1} \left[\sum_{i=1}^N z_i \alpha_i \right] \right. \\ & \left. + \left[T \sum_{i=1}^N z_i z_i' \right]^{-1} \left[\sum_{i=1}^N \sum_{t=1}^T z_i u_{it} \right] \right\} \xrightarrow{p} 0. \end{aligned} \quad (4.5)$$

It is easy to show that $\text{plim}_{T \rightarrow \infty} (1/N) \sum_{i=1}^N z_i \cdot (1/T) \sum_{t=1}^T u_{it} = 0$, but $(1/N) \sum_{i=1}^N z_i \alpha_i$ is a random variable with mean zero and covariance $\sigma^2 \sum_{i=1}^N z_i z_i' / N^2 \neq 0$ for finite N . Hence $\hat{\delta}_{MLE}$ is inconsistent when N is fixed and T tends to infinity. Similarly, using the argument of Amemiya (1967) we can show that $\hat{\lambda}_{MLE}$ and $\hat{\sigma}_{MLE}^2$ are not consistent in this case. Thus, we have the interesting situation that when N is fixed and T tends to infinity the model may be identifiable, yet δ , λ and σ^2 are not consistently estimable.

⁴For this and other technical details, see Anderson and Hsiao (1981b).

Intuitively, this is because if N is fixed we have a limited amount of information on α_i and z_i .

When T is fixed and N tends to infinity, we can show that the partial derivative equations divided by NT converge to zero at the true values and the information matrix is positive definite. By Amemiya's (1973) lemmas 3 and 4, we know that there is a consistent root for the MLE. The consistency of the MLE is a consequence of the fact that we are maximizing the likelihood function of N independently normally distributed T -component random vectors y_i .

The simultaneous solution of the first-order conditions is complicated. The standard Newton-Raphson iterative procedure for solving the MLE can be used. An alternative sequential iterative procedure may also be derived.

We note that from the first-order conditions we have

$$\begin{pmatrix} \beta \\ \delta \end{pmatrix} = \left[\sum_{i=1}^N \begin{pmatrix} y'_{i,-1} \\ z'_i e' \end{pmatrix} \left(I_T - \frac{1}{\lambda + T} e e' \right) \begin{pmatrix} y_{i,-1} & e z'_i \end{pmatrix} \right]^{-1} \\ \times \left[\sum_{i=1}^N \begin{pmatrix} y'_{i,-1} \\ z'_i e' \end{pmatrix} \left(I_T - \frac{1}{\lambda + T} e e' \right) y_i \right], \quad (4.6)$$

$$\sigma^2 = \frac{1}{NT\lambda} \left\{ \sum_{i=1}^N (y_i - y_{i,-1} \beta - e z'_i \delta)' \right. \\ \left. \times \left(I_T - \frac{1}{\lambda + T} e e' \right) (y_i - y_{i,-1} \beta - e z'_i \delta) \right\}, \quad (4.7)$$

$$\lambda = \frac{T \sum_{i=1}^N (\bar{y}_i - \bar{y}_{i,-1} \beta - \delta' z_i)^2}{N\sigma^2} - T. \quad (4.8)$$

Thus, we may use an iterative procedure to obtain the MLE by first substituting an initial trial value of λ into (4.6) to estimate β and δ and estimate σ^2 by (4.7) using the solution of (4.6). Substituting the solutions of (4.6) and (4.7) into (4.8) we obtain a revised estimate of λ . Then repeat the process by substituting the new value of λ into (4.6) to obtain a new estimate of β and δ , and so on until the solution converges.

This sequential iteration will converge to the MLE. Substituting (4.7) into (4.3) we have the concentrated log likelihood function

$$\log L = -\frac{NT}{2} (\log 2\pi + 1) - \frac{NT}{2} \log \sigma^2 - \frac{N(T-1)}{2} \log \lambda - \frac{N}{2} \log (\lambda + T).$$

Conditional on λ , (4.6) is the minimization solution of $\log \sigma^2$. Conditional on β and δ , (4.8) is the maximization solution of the concentrated likelihood function. The likelihood function is concave, therefore the iterative solution of (4.6)–(4.8) will converge.

For Case IIa, the log likelihood function is

$$\begin{aligned} \log L = & -\frac{NT}{2} \log 2\pi - \frac{NT}{2} \log \sigma^2 - \frac{N}{2} \log |A| - \frac{1}{2\sigma^2} \sum_{i=1}^N [(y_i - ez'_i \delta) \\ & - (y_{i,-1} - ez'_i \delta)\beta]' A^{-1} [(y_i - ez'_i \delta) - (y_{i,-1} - ez'_i \delta)\beta] \\ & - \frac{N}{2} \log 2\pi - \frac{N}{2} \log \sigma_e^2 - \frac{1}{2\sigma_e^2} \sum_{i=1}^N (y_{i0} - z'_i \delta)^2. \end{aligned} \quad (4.9)$$

For Case IIb, the log likelihood function is

$$\begin{aligned} \log L = & -\frac{NT}{2} \log 2\pi - \frac{NT}{2} \log \sigma_u^2 \\ & - \frac{1}{2\sigma_u^2} \sum_{i=1}^N \sum_{t=1}^T [(y_{it} - y_{i0}) - \beta(y_{i,t-1} - \tilde{\delta}' z_i)]^2 \\ & - \frac{N}{2} \log 2\pi - \frac{N}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_{i0} - \tilde{\delta}' z_i)^2. \end{aligned} \quad (4.10)$$

By an argument similar to that above, we can show that in Case IIa and Case IIb when N is fixed and T tends to infinity, the MLE of β is consistent, but the MLE of other parameters are inconsistent. When T is fixed and N tends to infinity, the MLE of all the parameters are consistent.

Taking the partial derivatives of (4.9) and setting them equal to zero, we have the following relationships:

$$\begin{aligned} \beta = & \left[\sum_{i=1}^N (y_{i,-1} - ez'_i \delta)' \left(I_T - \frac{1}{\lambda + T} ee' \right) (y_{i,-1} - ez'_i \delta) \right]^{-1} \\ & \times \sum_{i=1}^N (y_{i,-1} - ez'_i \delta)' \left(I_T - \frac{1}{\lambda + T} ee' \right) (y_i - ez'_i \delta), \end{aligned} \quad (4.11)$$

$$\begin{aligned} \sigma^2 = & \frac{1}{\lambda NT} \left\{ \sum_{i=1}^N [(y_i - ez'_i \delta) - (y_{i,-1} - ez'_i \delta)\beta]' \left(I_T - \frac{1}{\lambda + T} ee' \right) \right. \\ & \left. \times [(y_i - ez'_i \delta) - (y_{i,-1} - ez'_i \delta)\beta] \right\}, \end{aligned} \quad (4.12)$$

$$\begin{aligned} \tilde{\delta} = & \left\{ \left[T(1-\beta)^2 + \frac{(\lambda+T)\sigma^2}{\sigma_\varepsilon^2} \right] \sum_{i=1}^N z_i z_i' \right\}^{-1} \\ & \times \sum_{i=1}^N z_i \left[(\bar{y}_i - \bar{y}_{i-1} \beta)(1-\beta)T + \frac{(\lambda+T)\sigma^2}{\sigma_\varepsilon^2} y_{i0} \right], \end{aligned} \quad (4.13)$$

$$\lambda = \frac{T \sum_{i=1}^N [(\bar{y}_i - z_i' \tilde{\delta}) - \beta(\bar{y}_{i-1} - z_i' \tilde{\delta})]^2}{N\sigma^2} - T, \quad (4.14)$$

$$\sigma_\varepsilon^2 = \frac{1}{N} \sum_{i=1}^N (y_{i0} - z_i' \tilde{\delta})^2. \quad (4.15)$$

As one can see, each of (4.11)–(4.15) is a simple function of the other parameters. This suggests iterative procedure to obtain the MLE. We begin the iteration by noting that from the second part of the likelihood function (4.9), we can obtain an initial estimate of $\tilde{\delta}$ and σ_ε^2 by

$$\hat{\tilde{\delta}} = \left(\sum_{i=1}^N z_i z_i' \right)^{-1} \sum_{i=1}^N z_i y_{i0}, \quad (4.16)$$

and (4.15). Together with an initial value of λ , we substitute them into (4.11)–(4.15) sequentially so that at each stage we use the most recently estimated values to estimate other parameters.

Taking the partial derivatives of (4.10) and setting them equal to zero, we have

$$\beta = \left[\sum_{i=1}^N \sum_{t=1}^T (y_{i,t-1} - \delta' z_i)^2 \right]^{-1} \sum_{i=1}^N \sum_{t=1}^T (y_{i,t-1} - \tilde{\delta}' z_i)(y_{i,t} - y_{i0}), \quad (4.17)$$

$$\sigma_u^2 = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T [(y_{i,t} - y_{i0}) - \beta(y_{i,t-1} - \tilde{\delta}' z_i)]^2, \quad (4.18)$$

$$\begin{aligned} \tilde{\delta} = & \left\{ \left[\frac{T\beta}{\sigma_u^2} - \frac{1}{\sigma^2} \right] \left[\sum_{i=1}^N z_i z_i' \right] \right\}^{-1} \\ & \times \sum_{i=1}^N z_i \left[(\bar{y}_i - \beta \bar{y}_{i-1}) \frac{T\beta}{\sigma_u^2} + y_{i0} \left(\frac{T\beta}{\sigma_u^2} - \frac{1}{\sigma^2} \right) \right], \end{aligned} \quad (4.19)$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (y_{i0} - \tilde{\delta}' z_i)^2. \quad (4.20)$$

Therefore, we may obtain the MLE for Case IIb using (4.16) and (4.20) to start the iteration until the sequential solutions of (4.17)–(4.20) converge.

For Case III [eqs. (2.5) and (2.6) with w_{i0} fixed] the density of $(y_{i0}, \dots, y_{iT})$ is

$$\begin{aligned} f_i(y_{i0}, \dots, y_{iT}) &= f_i(y_{i1}, \dots, y_{iT} | y_{i0}) f_i(y_{i0}) \\ &= \left\{ \frac{1}{\sqrt{2\pi}\sigma_u} \right\}^T \exp \left\{ -\frac{1}{2\sigma_u^2} \sum_{t=1}^T [(y_{it} - y_{i0} + w_{i0}) \right. \\ &\quad \left. - \beta(y_{i,t-1} - y_{i0} + w_{i0})]^2 \right\} \\ &\quad \times \frac{1}{\sqrt{2\pi}\sigma_\eta} \exp \left\{ -\frac{1}{2\sigma_\eta^2} (y_{i0} - w_{i0} - \tilde{\delta}' z_i)^2 \right\}. \end{aligned} \quad (4.21)$$

For given y_{it} 's (4.21) is a function of w_{i0} 's, β , $\tilde{\delta}$, σ_u^2 and σ_η^2 . If we let $w_{i0} = y_{i0} - \tilde{\delta}' z_i$, then the second factor is $1/(\sqrt{2\pi}\sigma_\eta)$. As $\sigma_\eta \rightarrow 0$, this approaches ∞ and the likelihood function approaches ∞ . The likelihood function does not have a maximum and, hence, the MLE does not exist.

However, we can still take the partial derivatives of the joint likelihood function and solve for β , $\tilde{\delta}$, w_{i0} , σ_u^2 , σ_η^2 . By an argument similar to that of Anderson and Hsiao (1981a) we can show that when N is fixed and T tends to infinity the solution of β is equivalent to the CV, hence is consistent; but the solutions of other parameters are inconsistent. When T is fixed and N tends to infinity none of the coefficients are consistently estimable by solving the first derivative equations.

We may devise similar computational algorithms for solving the first derivative equations. However, the computation is much more complicated than in previous cases. However, when T is fixed, the estimator is inconsistent. When T goes to infinity, the estimator of β converges to the CV. Thus, it does not seem fruitful to elaborate on the computational algorithm for this case.

For Case IVa [eqs. (2.5) and (2.6) with w_{i0} stationary with variance $\lambda\sigma^2/(1-\beta^2)$], the density of $y_{i0}, \dots, y_{iT}$ is

$$\begin{aligned} f_i(y_{i0}, \dots, y_{iT}) &= (2\pi)^{-(T+1)/2} (\sigma^2)^{-(T+1)/2} |\Omega|^{-\frac{1}{2}} \\ &\quad \times \exp \left\{ -\frac{1}{2\sigma^2} [y_{i0} - \tilde{\delta}' z_i, (y_{i1} - \tilde{\delta}' z_i) - \beta(y_{i0} - \tilde{\delta}' z_i), \right. \\ &\quad \dots, (y_{iT} - \tilde{\delta}' z_i) - \beta(y_{i,T-1} - \tilde{\delta}' z_i)] \Omega^{-1} \\ &\quad \left. \times [y_{i0} - \tilde{\delta}' z_i, \dots, (y_{iT} - \tilde{\delta}' z_i) - \beta(y_{i,T-1} - \tilde{\delta}' z_i)] \right\}, \end{aligned} \quad (4.22)$$

where

$$\begin{aligned}\boldsymbol{\Omega} &= \lambda \begin{pmatrix} \frac{1}{1-\beta^2} & \boldsymbol{\theta}' \\ \boldsymbol{\theta} & \mathbf{I}_T \end{pmatrix} + \begin{pmatrix} \frac{1}{1-\beta} \\ \mathbf{e} \end{pmatrix} \begin{pmatrix} \frac{1}{1-\beta} & \mathbf{e}' \end{pmatrix}, \\ |\boldsymbol{\Omega}| &= \frac{\lambda^T}{1-\beta^2} \left[\lambda + T + \frac{1+\beta}{1-\beta} \right], \\ \boldsymbol{\Omega}^{-1} &= \frac{1}{\lambda} \left\{ \begin{pmatrix} 1-\beta^2 & \boldsymbol{\theta}' \\ \boldsymbol{\theta} & \mathbf{I}_T \end{pmatrix} - \begin{pmatrix} \lambda + T + \frac{1+\beta}{1-\beta} \end{pmatrix}^{-1} \begin{pmatrix} 1+\beta \\ \mathbf{e} \end{pmatrix} \begin{pmatrix} 1+\beta & \mathbf{e}' \end{pmatrix} \right\}.\end{aligned}\quad (4.23)$$

When N is fixed and T tends to infinity the MLE of β is consistent, but the MLE's of other parameters are inconsistent. The argument is similar to the usual one that the asymptotic theory is the same for both y_{i0} fixed and y_{i0} random. When T is fixed and N tends to infinity we are maximizing the joint density of N independently normally distributed $(T+1)$ -component random vectors $(y_{i0}, \dots, y_{iT})$; therefore the MLE's of all the parameters are consistent.

Taking partial derivatives of the logarithm of $\prod_{i=1}^N f_i(y_{i0}, \dots, y_{iT})$ with respect to σ^2 and $\boldsymbol{\delta}$ and setting them equal to zero, we have

$$\begin{aligned}\sigma^2 &= \frac{1}{N(T+1)\lambda} \left\{ \sum_{i=1}^N \left[(1-\beta^2)(y_{i0} - \boldsymbol{\delta}'\mathbf{z}_i)^2 \right. \right. \\ &\quad \left. \left. + \sum_{t=1}^T [(y_{it} - \boldsymbol{\delta}'\mathbf{z}_i) - \beta(y_{i,t-1} - \boldsymbol{\delta}'\mathbf{z}_i)]^2 \right] \right. \\ &\quad \left. - \frac{1-\beta}{2+(\lambda+T-1)(1-\beta)} \sum_{i=1}^N \left[(1+\beta)(y_{i0} - \boldsymbol{\delta}'\mathbf{z}_i) \right. \right. \\ &\quad \left. \left. + \sum_{t=1}^T [(y_{it} - \boldsymbol{\delta}'\mathbf{z}_i) - \beta(y_{i,t-1} - \boldsymbol{\delta}'\mathbf{z}_i)]^2 \right] \right\},\end{aligned}\quad (4.24)$$

$$\begin{aligned}\boldsymbol{\delta} &= \left\{ \left[(1-\beta^2) + T(1-\beta)^2 - \frac{[2+(T-1)(1-\beta)]^2(1-\beta)}{2+(\lambda+T-1)(1-\beta)} \right] \sum_{i=1}^N \mathbf{z}_i \mathbf{z}_i' \right\}^{-1} \\ &\quad \times \sum_{i=1}^N \mathbf{z}_i \frac{\lambda(1-\beta)^2}{2+(\lambda+T-1)(1-\beta)} [(1+\beta)y_{i0} + T(\bar{y}_i - \beta\bar{y}_{i-1})].\end{aligned}\quad (4.25)$$

Substituting (4.24) into the joint likelihood function and concentrating it, we have

$$\begin{aligned}\log L = & -[N(T+1)/2](\log 2\pi + 1) \\ & -[N(T+1)/2] \log \sigma^2 - (NT/2) \log \lambda \\ & - (N/2) \log [2 + (\lambda + T - 1)(1 - \beta)] + (N/2) \log (1 - \beta^2) \\ & + (N/2) \log (1 - \beta).\end{aligned}\quad (4.26)$$

Thus, we may obtain the MLE by the following procedure:⁵

Step 1. Estimate $\tilde{\delta}$ by

$$\hat{\tilde{\delta}} = \left(\sum_{i=1}^N \mathbf{z}_i \mathbf{z}_i' \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}_i' \bar{y}_i \right). \quad (4.27)$$

Step 2. Substitute (4.27) into (4.24) and (4.26) and search for the maximum value of (4.26) in the parameter space of β and λ $(-1, 1) \times (0, \infty)$.

Step 3. Substitute the solution of Step 2 into (4.25) and (4.24) to estimate $\tilde{\delta}$ and σ^2 .

The newly estimated $\tilde{\delta}$ may be substituted into Steps 2 and 3 again and continue the iteration until the solution converges.

We note that in the case that the MLE is consistent, there is the problem of boundary solutions [Anderson and Hsiao (1981a), Maddala (1971b), Nerlove (1971)]. Using the same method as Maddala (1971b) we can show that in all these cases except for Case IIb the boundary solution of $\sigma_u^2 = 0$ (or $\lambda = 0$) cannot occur, but the boundary solution of $\sigma_\eta^2 = 0$ (or $\lambda = \infty$) can occur. However, the probability of a boundary solution tends to zero as $T \rightarrow \infty$ or $N \rightarrow \infty$.

The iterative procedures for obtaining the MLE are complicated. Asymptotic efficiency in one iteration is achieved by using a consistent estimator as an initial estimator. The CV can only estimate β , and it is consistent only when T tends to infinity. We now suggest an instrumental variable method to obtain an initial estimator.⁶

Step 1. Premultiply (4.1) by the $(T-1) \times T$ transformation matrix that has -1 in the i th position and 1 in the $(i, i+1)$ th position $i = 1, \dots, T-1$, and 0's

⁵Lee (1979) has also suggested an iterative procedure similar to this one.

⁶The idea of using a variable which is purged of its dependence on the unobserved α_i as an instrument has also been suggested by Chamberlain (1977a,b) in a different context.

elsewhere to eliminate α_i and z_i , we have

$$y_{it} - y_{i,t-1} = \beta(y_{i,t-1} - y_{i,t-2}) + u_{it} - u_{i,t-1}. \quad (4.28)$$

Eq. (4.28) implies that we may use either $y_{i,t-2}$ or $(y_{i,t-2} - y_{i,t-3})$ as instruments. Thus we may estimate β by

$$\hat{\beta}_{IV} = \frac{\sum_{i=1}^N \sum_{t=3}^T (y_{it} - y_{i,t-1})(y_{i,t-2} - y_{i,t-3})}{\sum_{i=1}^N \sum_{t=3}^T (y_{i,t-1} - y_{i,t-2})(y_{i,t-2} - y_{i,t-3})}, \quad (4.29a)$$

or

$$\tilde{\beta}_{IV} = \frac{\sum_{i=1}^N \sum_{t=2}^T (y_{it} - y_{i,t-1}) y_{i,t-2}}{\sum_{i=1}^N \sum_{t=2}^T (y_{i,t-1} - y_{i,t-2}) y_{i,t-2}}. \quad (4.29b)$$

Step 2. Substitute $\hat{\beta}_{IV}$ or $\tilde{\beta}_{IV}$ into the following equation:

$$\bar{y}_i - \beta \bar{y}_{i,-1} = \delta' z_i + \alpha_i + \bar{u}_i, \quad i = 1, \dots, N, \quad (4.30)$$

and apply least squares method to estimate δ .

The instrumental variable method of estimating β is consistent either as T tends to infinity or as N tends to infinity, or both, and is independent of initial conditions. The least squares regression of $\bar{y}_i - \hat{\beta}_{IV} \bar{y}_{i,-1}$ or $\bar{y}_i - \tilde{\beta}_{IV} \bar{y}_{i,-1}$ on z_i will be consistent if T is fixed and N goes to infinity, but will not be consistent if N is fixed and T tends to infinity.

We may summarize the consistency properties of various estimators under various assumptions in table 1.

5. Models with time-varying exogenous variables — Serial correlation or state dependence

In this section we consider the formulation of dynamic models with time-varying exogenous variables, x_{it} . We note in section 2 [eqs. (2.3), (2.4), (2.5) and (2.6)] that we may write down the relationship between the unobserved process $\{w_{it}\}$ and the observed process $\{y_{it}\}$ either in the form

$$w_{it} = \beta w_{i,t-1} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (5.1)$$

$$y_{it} = w_{it} + \gamma' x_{it} + \eta_{it}, \quad i = 1, \dots, N, \quad t = 0, 1, \dots, T, \quad (5.2)$$

Table 1
Consistency properties of various estimators under various assumptions.^a

		MLE of		CV of		IV of	
		β	δ	β	δ	β	δ
Case I y_{10} fixed	N fixed	Consistent	Inconsistent	Consistent	Not estimable	Consistent	Inconsistent
	$T \rightarrow \infty$						
	T fixed	Consistent	Consistent	Inconsistent	Not estimable	Consistent	Consistent
Case II y_{10} random	$N \rightarrow \infty$						
	N fixed	Consistent	Inconsistent	Consistent	Not estimable	Consistent	Inconsistent
	$T \rightarrow \infty$						
Case III w_{10} fixed	T fixed	Consistent	Consistent	Inconsistent	Not estimable	Consistent	Consistent
	$N \rightarrow \infty$						
	N fixed	Consistent	Inconsistent	Consistent	Not estimable	Consistent	Inconsistent
Case IV w_{10} random	$T \rightarrow \infty$						
	T fixed	Inconsistent	Inconsistent	Inconsistent	Not estimable	Consistent	Consistent
	$N \rightarrow \infty$						
	N fixed	Consistent	Inconsistent	Consistent	Not estimable	Consistent	Inconsistent
	$T \rightarrow \infty$						
	T fixed	Consistent	Consistent	Inconsistent	Not estimable	Consistent	Consistent
	$N \rightarrow \infty$						
	N fixed	Consistent	Inconsistent	Consistent	Not estimable	Consistent	Inconsistent
	$T \rightarrow \infty$						
	T fixed	Consistent	Consistent	Inconsistent	Not estimable	Consistent	Consistent
	$N \rightarrow \infty$						

^aWhen the MLE does not exist, we replace it by the interior solution.

or in the form

$$w_{it} = \beta w_{i,t-1} + \gamma' x_{it} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (5.3)$$

$$y_{it} = w_{it} + \eta_i, \quad i = 1, \dots, N, \quad t = 0, 1, \dots, T. \quad (5.4)$$

Eq. (5.1) and (5.2) imply that

$$y_{it} = \beta y_{i,t-1} + \gamma' x_{it} - \beta \gamma' x_{i,t-1} + \alpha_i + u_{it}. \quad (5.5)$$

Eq. (5.3) and (5.4) imply that

$$y_{it} = \beta y_{i,t-1} + \gamma' x_{it} + \alpha_i + u_{it}. \quad (5.6)$$

We shall call the former model the *serial correlation model* and the latter model the *state dependence model*.

In either case the unobservable variables w_{it} are serially correlated; the impact of a disturbance u_{it} tends to persist for more than one period [Chamberlain (1978) and Heckman (1978)]. In the serial correlation model y_{it} is only affected by x_{it} , not by $x_{i,t-1}, \dots$. In the state dependence model y_{it} is affected not merely by x_{it} , but by $x_{i,t-1}, \dots$ through $y_{i,t-1}$; the unobservable w_{it} carry the effected x and can be called a state. Inclusion of the time-invariant individual affect η_i implies the aggregate effect of unobservable variables (commonly called 'residual') are serially correlated in another way.

The key distinction between these two models is whether there is a dynamic response to changes in observed variables. In a serial correlation model if x is increased in period t and then returned to its former level, the distribution of y in period $t+1$ is not affected. Past y is informative because it helps to predict the effect of unobservable variables which are serially correlated. In the state dependence model a change in x that affects the distribution of y in period t will continue to affect the distribution of y in period $t+1$, even though the intervention is present only in period t .

As one can see, the serial correlation model [eqs. (5.1) and (5.2)] implies that y_{it} fluctuate around the equilibrium level $\gamma' x_{it} + \eta_i$. It is the effect of unobservable variables $\{w_{it}\}$ that follows a first-order autoregressive process. On the other hand, by continuous substitution of the state dependence model (5.6) both past and present x determine the current mean level of y_{it} .

Eqs. (5.5) and (5.6) imply a simple test between serial correlation and the state dependence model. With serial correlation lagged x enters and their coefficients are equal to minus the product of the coefficients of current x and lagged y . For models with time-invariant exogenous variables only (section 2) the current and lagged exogenous variables are identical; we cannot distinguish these two cases based on observed values of y and z only.

Again, with different assumptions about the initial observations we may have different models. For *serial correlation* models [eqs. (5.1) and (5.2)], it is relatively simple. The standard assumption is that either w_{i0} is fixed or w_{i0} is stationary with mean zero and variance $\sigma_u^2/(1-\beta^2)$.

For *state dependence* models (5.6), it is more complicated. Corresponding to the classifications we made in section 2, we classify it into four basic cases: y_{i0} fixed; y_{i0} random; w_{i0} fixed; and w_{i0} random.

Case I: y_{i0} fixed. In this case a cross-sectional unit may start at some arbitrary position y_{i0} and gradually move towards a level determined by $\eta_i + \gamma' \sum_{j=0}^{t-1} x_{i,t-j} \beta^j$. The individual effect, η_i , does not affect the initial y_{i0} , but affects all later y_{it} 's.

Case II: y_{i0} random. We assume y_{i0} are normally distributed with mean μ_{y_0} and variance $\sigma_{y_0}^2$. Depending on the assumption with regard to the correlation between y_{i0} and η_i , we divide this case into two sub-cases.

Iia: y_{i0} independent of η_i .

Iib: y_{i0} correlated with η_i . We denote the covariance between y_{i0} and η_i by $\rho\sigma_{y_0}^2$.

A rationalization of this assumption is that we may treat y_{it} as a state. We do not care how the initial state is reached as long as we know that it is normally distributed with finite mean and variance. We may let the initial state be independent of the equilibrium level (Iia) or help to determine the equilibrium level (Iib).

Case III: w_{i0} fixed. The unobserved individual process $\{w_{it}\}$ is independent of the individual effect η_i with the starting value arbitrary. Or each of the observed cross-sectional units may start at some arbitrary position y_{i0} and gradually move towards a level of $\eta_i + \gamma' x_{i,t} + \beta\gamma' x_{i,t-1} + \beta^2\gamma' x_{i,t-2} + \dots$ with the individual η_i affecting the starting value y_{i0} .

Case IV: w_{i0} random. Given the unobservable process $\{w_{it}\}$ and the individual effects η_i are independent, we may formulate the initial state w_{i0} in several different ways:

IVa: w_{i0} random with common mean μ_w and variance $\sigma_u^2/(1-\beta^2)$.

IVb: w_{i0} random with common mean μ_w and variance $\sigma_{w_0}^2$.

IVc: w_{i0} random with mean θ_{i0} and variance $\sigma_u^2/(1-\beta^2)$.

IVd: w_{i0} random with mean θ_{i0} and variance $\sigma_{w_0}^2$.

IVa and IVb assume that the initial state is a random draw from a distribution with finite mean. IVa assumes that the initial state has the same variance as the latter states. IVb allows the initial state to be non-stationary (with arbitrary variance). IVc and IVd assume that the individual states are random draws from different populations with different means. A rationalization of this is that by successive substitution of (5.6) we have

$$w_{it} = \gamma' \sum_{j=0}^{\infty} x_{i,t-j} \beta^j + u_{it} + \beta u_{i,t-1} + \beta^2 u_{i,t-2} + \cdots \quad (5.7)$$

When $t=0$ we have

$$w_{i0} = \gamma' \sum_{j=0}^{\infty} x_{i,-j} \beta^j + u_{i0} + \beta u_{i,-1} + \beta^2 u_{i,-2} + \cdots \quad (5.8)$$

Since $x_0, x_{-1}, \dots$ are not observable, we may treat the cumulative effect of past x 's as an unknown parameter and let

$$\theta_{i0} = \gamma' \sum_{j=0}^{\infty} x_{i,-j} \beta^j. \quad (5.9)$$

IVc assumes that the process $\{w_{it}\}$ was generated from the infinite past and has achieved stationarity after conditioning the exogenous variables. IVd relaxes this assumption by allowing the variance of w_{i0} to be arbitrary.

6. Estimation of models with serial correlation

In this section we consider estimation methods for a regression model whose disturbance is additive in two independent components, one of which is an unobservable individual effect $\alpha_i = (1 - \beta)\eta_i$ and the other, w_{it} , follows a first-order autoregressive process. Estimation of models with composite disturbance terms have been widely discussed in the recent literature [e.g., Pagan (1973), Pagano (1974), Revankar (1980), etc.]. However, most authors assume both components are time varying. Here, the components are independently distributed across cross-sectional units, but one is time-invariant and the other follows an autoregressive process. The problem provides an interesting generalization of the standard regression model with serially correlated residuals — interesting because in this particular circumstance it lends itself to ready comparison with the standard estimation procedures.

To be consistent with the rest of the paper, we assume in this section that sample observations start at time 0 so that both y_{i0} and x_{i0} are observable. When w_{i0} is fixed, the joint density of $(y_{i0}, \dots, y_{iT})$ is

$$\begin{aligned}
 f_i(y_{i0}, \dots, y_{iT}) = & (2\pi)^{-T/2} (\sigma_u^2)^{-T/2} \exp \left\{ -\frac{1}{2\sigma_u^2} \sum_{t=1}^T [(y_{it} - y_{i0} + w_{i0}) \right. \\
 & - \beta(y_{i,t-1} - y_{i0} + w_{i0}) - \gamma'(\mathbf{x}_{it} - \mathbf{x}_{i0}) \\
 & \left. - \gamma'(\mathbf{x}_{i,t-1} - \mathbf{x}_{i0})\beta]^2 \right\} (2\pi)^{-\frac{1}{2}} (\sigma_\eta^2)^{-\frac{1}{2}} \\
 & \times \exp \left\{ -\frac{1}{2\sigma_\eta^2} (y_{i0} - \gamma' \mathbf{x}_{i0} - w_{i0})^2 \right\}. \quad (6.1)
 \end{aligned}$$

By letting $w_{i0} = y_{i0} - \gamma' \mathbf{x}_{i0}$ and $\sigma_\eta^2 \rightarrow 0$, we see the likelihood function is unbounded. Thus, contrary to the standard model with serially correlated residuals, here the MLE does not exist when w_{i0} is fixed.

We may still take partial derivatives of the joint likelihood function and set them equal to zero. We obtain the following relationships:

$$\begin{aligned}
 \beta = & \left(\sum_{i=1}^N \sum_{t=1}^T [(y_{it} - y_{i0} + w_{i0}) - \gamma'(\mathbf{x}_{it} - \mathbf{x}_{i0})] \right. \\
 & \times [(y_{i,t-1} - y_{i0} + w_{i0}) - \gamma'(\mathbf{x}_{i,t-1} - \mathbf{x}_{i0})] \Big) \\
 & \left/ \left(\sum_{i=1}^N \sum_{t=1}^T [(y_{i,t-1} - y_{i0} + w_{i0}) - \gamma'(\mathbf{x}_{i,t-1} - \mathbf{x}_{i0})]^2 \right) \right), \quad (6.2)
 \end{aligned}$$

$$\begin{aligned}
 \sigma_u^2 = & \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \left\{ (y_{it} - y_{i0} + w_{i0}) - \beta(y_{i,t-1} - y_{i0} + w_{i0}) \right. \\
 & \left. - \gamma'[(\mathbf{x}_{it} - \mathbf{x}_{i0}) - (\mathbf{x}_{i,t-1} - \mathbf{x}_{i0})\beta] \right\}^2, \quad (6.3)
 \end{aligned}$$

$$\sigma_\eta^2 = \frac{1}{N} \sum_{i=1}^N (y_{i0} - \gamma' \mathbf{x}_{i0} - w_{i0})^2, \quad (6.4)$$

$$\begin{aligned}
\gamma = & \left\{ \frac{1}{\sigma_u^2} \sum_{i=1}^N \sum_{t=1}^T [(x_{it} - x_{i0}) - (x_{i,t-1} - x_{i0})\beta] \right. \\
& \times [(x_{i,t} - x_{i0}) - (x_{i,t-1} - x_{i0})\beta]' + \frac{1}{\sigma_\eta^2} \sum_{i=1}^N x_{i0} x_{i0}' \left. \right\}^{-1} \\
& \times \left\{ \frac{1}{\sigma_u^2} \sum_{i=1}^N \sum_{t=1}^T [(x_{it} - x_{i0}) - (x_{i,t-1} - x_{i0})\beta] \right. \\
& \times [(y_{it} - y_{i0} + w_{i0}) - \beta(y_{i,t-1} - y_{i0} + w_{i0})] + \frac{1}{\sigma_\eta^2} \sum_{i=1}^N x_{i0}(y_{i0} - w_{i0}) \left. \right\},
\end{aligned} \tag{6.5}$$

$$w_{i0} = (y_{i0} - \gamma' x_{i0}) - \frac{T(1-\beta)\sigma_\eta^2}{T(1-\beta)^2\sigma_\eta^2 + \sigma_u^2} [\bar{y}_i - \bar{y}_{i,-1}\beta - \gamma'(\bar{x}_i - \bar{x}_{i,-1}\beta)], \tag{6.6}$$

where

$$\bar{x}_i = \frac{1}{T} \sum_{t=1}^T x_{it}, \quad \bar{x}_{i,-1} = \frac{1}{T} \sum_{t=1}^T x_{i,t-1}.$$

We note that when T tends to infinity, (6.6) converges to

$$\hat{w}_{i0} = (y_{i0} - \bar{y}_i) - \gamma'(x_{i0} - \bar{x}). \tag{6.7}$$

Therefore we may suggest the following iterative procedure to solve for the MLE:

Step 1. Obtain initial values of γ and w_{i0} . An initial value of γ may be obtained by

$$\hat{\gamma}_{OLS} = \left(\sum_{i=1}^N \sum_{t=1}^T x_{it} x_{it}' \right)^{-1} \sum_{i=1}^N \sum_{t=1}^T x_{it} y_{it}, \tag{6.8}$$

and w_{i0} by substituting (6.8) into (6.7).

Step 2. Sequentially solve (6.2)–(6.6) using (6.7) and (6.8) as the initial values. Repeat the sequential iteration using the most recently estimated value until the solution converges.

We note that substituting (6.6) into the likelihood function (6.1) and taking partial derivatives of the concentrated log likelihood function with respect to

γ and setting them equal to zero, we can show that when either T or N tends to infinity, $\hat{\gamma}$ converges to γ independent of whether we know the true values of β , σ_u^2 and σ_η^2 or not.⁷ Thus the interior solution of γ is consistent.

Substituting (6.6) and $\hat{\gamma}$ into (6.2) and using the fact that $\hat{\gamma}$ converges to γ and the similar argument of Anderson and Hsiao (1981a), we can show that the solution of β is consistent when N is fixed and T tends to infinity and inconsistent when T is fixed and N tends to infinity. As far as the solutions for w_{i0} , σ_u^2 and σ_η^2 are concerned, it is easy to show that they are inconsistent when either T is fixed and N tends to infinity or N is fixed and T tends to infinity.

When w_{i0} is normally distributed with mean zero and variance $\lambda\sigma^2/(1-\beta^2)$, the joint density of $(y_{i0}, \dots, y_{iT})$ is

$$\begin{aligned} f(y_{i0}, \dots, y_{iT}) = & (2\pi)^{-(T+1)/2} (\sigma^2)^{-(T+1)/2} |\Omega|^{-\frac{1}{2}} \\ & \times \exp \left\{ -\frac{1}{2\sigma^2} [y_{i0} - \gamma' x_{i0}, y_{i1} - \beta y_{i0} \right. \\ & - \gamma'(x_{i1} - x_{i0}\beta), \dots, y_{iT} - \beta y_{i, T-1} \\ & \left. - \gamma'(x_{iT} - x_{i, T-1}\beta)] \Omega^{-1} \right. \\ & \left. \times [y_{i0} - \gamma' x_{i0}, \dots, y_{iT} - \beta y_{i, T-1} - \gamma'(x_{iT} - x_{i, T-1}\beta)]' \right\}. \end{aligned} \quad (6.9)$$

By the same reasoning as that discussed for Case IVa in section 3, we know that the MLE in this case is consistent when either T or N tends to infinity.

To obtain the MLE we take partial derivatives of the logarithm of the joint density function (6.9) with respect to σ^2 and γ and set them equal to zero; we have

$$\begin{aligned} \sigma^2 = & \frac{1}{N(T+1)} \left\{ \sum_{i=1}^N \left[(1-\beta^2)(y_{i0} - \gamma' x_{i0})^2 \right. \right. \\ & \left. \left. + \sum_{t=1}^T (y_{it} - \beta y_{i, t-1} - \gamma' x_{it} + \gamma' x_{i, t-1}\beta)^2 \right] \right. \\ & \left. - \frac{1-\beta}{2+(\lambda+T-1)(1-\beta)} \sum_{i=1}^N \left[(1+\beta)(y_{i0} - \gamma' x_{i0}) \right. \right. \\ & \left. \left. + \sum_{t=1}^T (y_{it} - \beta y_{i, t-1} - \gamma' x_{it} + \gamma' x_{i, t-1}\beta) \right]^2 \right\}, \end{aligned} \quad (6.10)$$

⁷For detail, see Anderson and Hsiao (1981b).

$$\begin{aligned}
\gamma = & \left\{ (1 - \beta^2) \sum_{i=1}^N \mathbf{x}_{i0} \mathbf{x}'_{i0} + \sum_{i=1}^N \sum_{t=1}^T (\mathbf{x}_{it} - \mathbf{x}_{i,t-1} \beta) (\mathbf{x}_{it} - \mathbf{x}_{i,t-1} \beta)' \right. \\
& - \frac{1 - \beta}{2 + (\lambda + T - 1)(1 - \beta)} \sum_{i=1}^N \left[(1 + \beta) \mathbf{x}_{i0} + \sum_{t=1}^T (\mathbf{x}_{it} - \mathbf{x}_{i,t-1} \beta) \right] \\
& \times \left[(1 + \beta) \mathbf{x}_{i0} + \sum_{t=1}^T (\mathbf{x}_{it} - \mathbf{x}_{i,t-1} \beta) \right]^{-1} \\
& \times \left\{ (1 - \beta^2) \sum_{i=1}^N \mathbf{x}_{i0} y_{i0} + \sum_{i=1}^N \sum_{t=1}^T (\mathbf{x}_{it} - \mathbf{x}_{i,t-1} \beta) (y_{it} - \beta y_{i,t-1}) \right. \\
& - \frac{1 - \beta}{2 + (\lambda + T - 1)(1 - \beta)} \sum_{i=1}^N \left[(1 + \beta) \mathbf{x}_{i0} + \sum_{t=1}^T (\mathbf{x}_{it} - \mathbf{x}_{i,t-1} \beta) \right] \\
& \times \left. \left. \left[(1 + \beta) y_{i0} + \sum_{t=1}^T (y_{it} - \beta y_{i,t-1}) \right] \right\}. \quad (6.11)
\end{aligned}$$

Substituting (6.10) into the joint likelihood function, we obtain the concentrated log likelihood function of the form (4.26). For given γ we can find the maximum of (4.26) by searching on the parameter space of β and λ $(-1, 1) \times (0, \infty)$. Therefore we may use the least squares estimator of γ , (6.8), as the initial value to begin the search. The values of β and λ which maximize (4.26) conditional on $\hat{\gamma}_{OLS}$, are then substituted into (6.10) and (6.11) to get new estimates of γ and σ^2 . Conditional on the revised estimate of γ , we again search for the maximum of (4.26) on the parameter space of β and λ and so on until the solution converges. We summarize the consistency properties in table 2.

Table 2
The consistency property of the MLE for the serial correlation model.

		N fixed $T \rightarrow \infty$	T fixed $N \rightarrow \infty$
w_{i0} fixed	γ	Consistent	Consistent
	β	Consistent	Inconsistent
	$\sigma_u^2, \sigma_\eta^2, w_{i0}$	Inconsistent	Inconsistent
w_{i0} stationary	γ, β	Consistent	Consistent
	$\sigma_u^2, \sigma_\eta^2$	Inconsistent	Consistent

As we have seen, the MLE is very complicated whether w_{i0} is assumed fixed or random. However, the information matrix of γ and β , σ_u^2 , and σ_η^2 is block-diagonal. That is, the information about the covariance matrix of residuals is asymptotically irrelevant. Thus, the generalized least squares using the true variance-covariance matrix and the generalized least squares using the estimated variance covariance matrix are asymptotically equivalent [Maddala (1971a)]. Hence, we may use the following multi-step estimation procedure to estimate γ .

Step 1. Eliminate individual η_i by subtracting individual mean from (5.5). We have

$$(y_{it} - \bar{y}_i) = \gamma'(\mathbf{x}_{it} - \bar{\mathbf{x}}_i) + (w_{it} - \bar{w}_i), \quad (6.12)$$

where $\bar{w}_i = (1/T) \sum_{t=1}^T w_{it}$.

Step 2. Obtain a consistent estimate of β by applying standard procedure to (6.12). For instance, we may use the Durbin (1960) method by regressing $(y_{it} - \bar{y}_i)$ on $(y_{i,t-1} - \bar{y}_{i,-1})$, $(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)$, and $(\mathbf{x}_{i,t-1} - \bar{\mathbf{x}}_{i,-1})$ and treat the coefficient of $(y_{i,t-1} - \bar{y}_{i,-1})$ as the estimated value of β . Or we may use least squares residuals of (6.12) and estimate β by

$$\hat{\beta} = \left(\sum_{i=1}^N \sum_{t=1}^T \hat{w}_{it} \hat{w}_{i,t-1} \right) / \left(\sum_{i=1}^N \sum_{t=1}^T \hat{w}_{i,t-1}^2 \right), \quad (6.13)$$

where $\hat{w}_{it} = (y_{it} - \bar{y}_i) - \hat{\gamma}'_{OLS}(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)$.

Step 3. Estimate σ_u^2 , σ^2 , and λ by

$$\begin{aligned} \hat{\sigma}_u^2 = & \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \left\{ (y_{it} - \bar{y}_i) - \hat{\beta}(y_{i,t-1} - \bar{y}_{i,-1}) \right. \\ & \left. - \hat{\gamma}'_{OLS}[(\mathbf{x}_{it} - \bar{\mathbf{x}}_i) - (\mathbf{x}_{i,t-1} - \bar{\mathbf{x}}_{i,-1}) \hat{\beta}] \right\}^2, \end{aligned} \quad (6.14)$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N [\bar{y}_i - \hat{\beta} \bar{y}_{i,-1} - \hat{\gamma}'_{OLS}(\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_{i,-1}) \hat{\beta}]^2 - \frac{1}{T} \hat{\sigma}_u^2, \quad (6.15)$$

$$\hat{\lambda} = \hat{\sigma}_u^2 / \hat{\sigma}^2. \quad (6.16)$$

Step 4. Estimate γ by

$$\hat{\gamma} = \left[\sum_{i=1}^N (X_i - \hat{\beta} X_{i,-1})' \left(I_T - \frac{1}{\hat{\lambda} + T} ee' \right) (X_i - \hat{\beta} X_{i,-1}) \right]^{-1} \times \left[\sum_{i=1}^N (X_i - \hat{\beta} X_{i,-1})' \left(I_T - \frac{1}{\hat{\lambda} + T} ee' \right) (y_i - \hat{\beta} y_{i,-1}) \right], \quad (6.17)$$

where $X_{i,-1} = (x_{i0}, x_{i1}, \dots, x_{iT-1})'$.

When T tends to infinity, $\hat{\lambda}$ is asymptotically irrelevant, since $\hat{\lambda}/\sqrt{T} \rightarrow 0$ in probability. We may replace Steps 3 and 4 by:

Step 3'. Estimate γ by

$$\begin{aligned} \hat{\gamma} = & \left\{ \sum_{i=1}^N \sum_{t=1}^T [(x_{it} - \bar{x}_i) - (x_{i,t-1} - \bar{x}_{i,-1})\hat{\beta}] \right. \\ & \times [(x_{it} - \bar{x}_i) - (x_{i,t-1} - \bar{x}_{i,-1})\hat{\beta}]' \left. \right\}^{-1} \\ & \times \left\{ \sum_{i=1}^N \sum_{t=1}^T [(x_{it} - \bar{x}_i) - (x_{i,t-1} - \bar{x}_{i,-1})\hat{\beta}] \right. \\ & \times [(y_{it} - \bar{y}_i) - (y_{i,t-1} - \bar{y}_{i,-1})\hat{\beta}]' \left. \right\}. \end{aligned} \quad (6.18)$$

Estimator (6.18) is asymptotically efficient when T tends to infinity.

7. Estimation of models with state dependence

In this section we consider estimation methods for models with state dependence. We make the usual standard assumption that

$$\text{rank} \left[\frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T x_{it} x_{it}' \right] = K_2, \quad (7.1)$$

and $1/(NT) \sum_{i=1}^N \sum_{t=1}^T x_{it} x_{it}'$ converges to a nonsingular constant matrix as $T \rightarrow \infty$ or $N \rightarrow \infty$ or both. Using a similar argument as was used in section 3, we know that (7.1) is a necessary and sufficient condition for the

identification of model (5.6). Since, except for the ill-behaved x , this condition in general will be satisfied for reasonable N or T , we shall assume model (5.6) is always identified in the discussion.

We first consider the property of the CV. The CV's for β and γ are

$$\begin{aligned} \hat{\beta}_{CV} = & \left[\sum_{i=1}^N y'_{i,-1} Q y_{i,-1} \right. \\ & - \left(\sum_{i=1}^N y'_{i,-1} Q X_i \right) \left(\sum_{i=1}^N X_i Q X_i \right)^{-1} \left(\sum_{i=1}^N X_i Q y_{i,-1} \right) \Big]^{-1} \\ & \times \left[\sum_{i=1}^N y'_{i,-1} Q y_i \right. \\ & \left. - \left(\sum_{i=1}^N y'_{i,-1} Q X_i \right) \left(\sum_{i=1}^N X_i Q X_i \right)^{-1} \left(\sum_{i=1}^N X_i Q y_i \right) \right], \end{aligned} \quad (7.2)$$

$$\begin{aligned} \hat{\gamma}_{CV} = & \left[\sum_{i=1}^N X_i Q X_i \right. \\ & - \left(\sum_{i=1}^N X_i Q y_{i,-1} \right) \left(\sum_{i=1}^N y'_{i,-1} Q y_{i,-1} \right)^{-1} \left(\sum_{i=1}^N y'_{i,-1} Q X_i \right) \Big]^{-1} \\ & \times \left[\sum_{i=1}^N X_i Q y_i \right. \\ & \left. - \left(\sum_{i=1}^N X_i Q y_{i,-1} \right) \left(\sum_{i=1}^N y'_{i,-1} Q y_{i,-1} \right)^{-1} \left(\sum_{i=1}^N y'_{i,-1} Q y_i \right) \right], \end{aligned} \quad (7.3)$$

where $X_i = (x_{i1}, \dots, x_{iT})'$.

It is easy to show that the probability limits of the inverse matrices [in (7.2) and (7.3)] divided by NT are non-singular constant matrices, and that

$$\text{plim} \frac{1}{NT} \sum_{i=1}^N X_i Q u_i \quad (7.4)$$

is zero when T tends to infinity or N tends to infinity, or both. We also know by a law of large numbers that

$$\text{plim}_{\substack{T \rightarrow \infty \\ N \text{ fixed}}} \frac{1}{NT} \sum_{i=1}^N y'_{i,-1} \mathbf{Q} u_i = 0, \quad (7.5)$$

$$\text{plim}_{\substack{N \rightarrow \infty \\ T \text{ fixed}}} \frac{1}{NT} \sum_{i=1}^N y'_{i,-1} \mathbf{Q} u_i = -\frac{\lambda \sigma^2}{(1-\beta)^2 T^2} [(T-1) - T\beta + \beta^T]. \quad (7.6)$$

Therefore, when N is fixed and T tends to infinity the CV of β and γ are consistent. Furthermore, by an argument similar to that of Anderson (1971, ch. 5), we can show that the CV is also asymptotically normally distributed as T tends to infinity. When T is fixed and N tends to infinity, the CV is inconsistent.

Another commonly used estimation method is the MLE. However, the property of the MLE not only depends on the way in which the cross-sectional units N or the time series observations T go to infinity, but also depends on the initial conditions. We shall now consider the consistency property of the MLE case by case.

For Case I where y_{i0} is fixed, the log likelihood function is

$$\begin{aligned} \log L = & -(NT/2) \log 2\pi - (NT/2) \log \sigma^2 - (N/2) \log |\mathbf{A}| \\ & - \frac{1}{2\sigma^2} \sum_{i=1}^N (\mathbf{y}_i - \mathbf{y}_{i,-1} \beta - \mathbf{X}_i \gamma)' \mathbf{A}^{-1} (\mathbf{y}_i - \mathbf{y}_{i,-1} \beta - \mathbf{X}_i \gamma), \end{aligned} \quad (7.7)$$

which is basically the same form as (4.3). Using (4.4) we can show that the MLE of β and γ converge to the CV [eqs. (7.2) and (7.3)] when T tends to infinity, which is consistent. But with the same consideration as Amemiya (1967), we know that λ and σ^2 are not consistently estimable when N is fixed.

When T is fixed and N tends to infinity, the partial derivative equations of (7.7) divided by NT converge to zero at their true values. The second derivative matrix divided by NT converges in probability to a negative definite matrix when $T \geq 2$. Therefore the MLE of β , γ , σ^2 , and λ are consistent. The MLE's are also asymptotically normally distributed.

A sequential algorithm can be used to obtain the MLE. We note that taking partial derivatives of (7.7) and setting them equal to zero, we have

$$\begin{aligned} \begin{pmatrix} \beta \\ \gamma \end{pmatrix} = & \left[\sum_{i=1}^N \begin{pmatrix} y_{i,-1} \\ \mathbf{X}_i \end{pmatrix} \left(\mathbf{I}_T - \frac{1}{\lambda + T} \mathbf{e} \mathbf{e}' \right) (\mathbf{y}_{i,-1} \mathbf{X}_i) \right]^{-1} \\ & \times \left[\sum_{i=1}^N \begin{pmatrix} y'_{i,-1} \\ \mathbf{X}'_i \end{pmatrix} \left(\mathbf{I}_T - \frac{1}{\lambda + T} \mathbf{e} \mathbf{e}' \right) \mathbf{y}_i \right], \end{aligned} \quad (7.8)$$

$$\sigma^2 = \frac{1}{NT\lambda} \left[\sum_{i=1}^N (y_i - y_{i-1} \beta - X_i \gamma)' \times \left(I_T - \frac{1}{\lambda + T} ee' \right) (y_i - y_{i-1} \beta - X_i \gamma) \right], \quad (7.9)$$

$$\lambda = \frac{1}{N\sigma^2} \left(T \sum_{i=1}^N (\bar{y}_i - \bar{y}_{i-1} \beta - \gamma' \bar{x}_i)^2 \right) - T. \quad (7.10)$$

Thus, we may obtain the MLE by first substituting an initial trial value of λ into (7.8) to estimate β and γ , and substitute the solution of (7.8) in (7.9) to estimate σ^2 . The solutions of (7.8) and (7.9) are substituted into (7.10) to get a revised estimated value of λ , which may be substituted back into (7.8) and (7.9) to obtain revised estimates of β and γ , and so on until the solution converges.

For Case IIa, the log likelihood function is

$$\begin{aligned} & -\frac{NT}{2} \log 2\pi - \frac{NT}{2} \log \sigma^2 - \frac{N}{2} \log |A| \\ & - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - y_{i-1} \beta - X_i \gamma)' A^{-1} (y_i - y_{i-1} \beta - X_i \gamma) \\ & - \frac{N}{2} \log 2\pi - \frac{N}{2} \log \sigma_0^2 - \frac{1}{2\sigma_0^2} \sum_{i=1}^N (y_{i0} - \mu_{y0})^2 \end{aligned} \quad (7.11)$$

Therefore, the method and the statistical properties of the MLE of β , γ , σ^2 and λ are identical to the case where y_{i0} are fixed constants, except that now in addition to estimating these parameters we also estimate μ_{y0} and σ_0^2 by

$$\hat{\mu}_{y0} = \frac{1}{N} \sum_{i=1}^N y_{i0}, \quad \hat{\sigma}_0^2 = \frac{1}{N} \sum_{i=1}^N (y_{i0} - \hat{\mu}_{y0})^2. \quad (7.12)$$

It is obvious that $\hat{\mu}_{y0}$ and $\hat{\sigma}_0^2$ are consistent when N tends to infinity but inconsistent when N is fixed and T tends to infinity.

For Case IIb, the log likelihood function is

$$\begin{aligned}
\log L = & -\frac{NT}{2} \log 2\pi - \frac{N(T-1)}{2} \log \sigma_u^2 - \frac{N}{2} \log(\sigma_u^2 + Ta) \\
& - \frac{1}{2\sigma_u^2} \sum_{i=1}^N \sum_{t=1}^T [y_{it} - \beta y_{i,t-1} - \gamma' \mathbf{x}_{it} - \rho(y_{i0} - \mu_{y0})]^2 \\
& + \frac{a}{2\sigma_u^2(\sigma_u^2 + Ta)} \sum_{i=1}^N \left\{ \sum_{t=1}^T [y_{it} - \beta y_{i,t-1} - \gamma' \mathbf{x}_{it} - \rho(y_{i0} - \mu_{y0})] \right\}^2 \\
& - \left\{ \frac{N}{2} \log 2\pi + \frac{N}{2} \log \sigma_{y0}^2 + \frac{1}{2\sigma_{y0}^2} \sum_{i=1}^N (y_{i0} - \mu_{y0})^2 \right\}, \quad (7.13)
\end{aligned}$$

where $a = \sigma^2 - \rho^2 \sigma_{y0}^2$. When N is fixed and T tends to infinity, (7.13) is dominated by the terms except the last one, which is basically the same form as (4.3) or (7.7) with $(y_{i0} - \mu_{y0})$ playing the role of the time-invariant variable z_i . Thus, by the same manipulation as before, we can show that the MLE of β and γ converge to the CV, hence are consistent; but the MLE of μ_{y0} , ρ , σ_u^2 and σ^2 converge to random variables, which are inconsistent. When T is fixed and N tends to infinity, Goodrich and Caines (1979) have shown that the MLE is consistent.

Taking partial derivatives and setting them equal to zero, we have

$$\begin{aligned}
\begin{pmatrix} \beta \\ \gamma \\ \rho \end{pmatrix} = & \sum_{i=1}^N \begin{pmatrix} y'_{i,-1} \\ X'_i \\ (y_{i0} - \mu_{y0})e' \end{pmatrix} \left(I_T - \frac{a}{\sigma_u^2 + Ta} ee' \right) \begin{pmatrix} y_{i,-1} \\ X_i \\ (y_{i0} - \mu_{y0})e \end{pmatrix} \Bigg\}^{-1} \\
& \times \sum_{i=1}^N (y_{i,-1}, X_i, e(y_{i0} - \mu_{y0}))' \left(I_T - \frac{a}{\sigma_u^2 + Ta} ee' \right) y_i, \quad (7.14)
\end{aligned}$$

$$\begin{aligned}
\sigma_u^2 = & \frac{1}{NT} \sum_{i=1}^N [y_i - y_{i,-1} \beta - X_i \gamma - \rho e(y_{i0} - \mu_{y0})]' \\
& \times \left(I_T - \frac{a}{\sigma_u^2 + Ta} ee' \right) [y_i - y_{i,-1} \beta - X_i \gamma - \rho e(y_{i0} - \mu_{y0})], \quad (7.15)
\end{aligned}$$

$$a = \frac{1}{N} \sum_{i=1}^N [\bar{y}_i - \bar{y}_{i,-1} \beta - \bar{x}'_i \gamma - \rho(y_{i0} - \mu_{y0})]^2 - \frac{1}{T} \sigma_u^2, \quad (7.16)$$

$$\sigma_{y0}^2 = \frac{1}{N} \sum_{i=1}^N (y_{i0} - \mu_{y0})^2, \quad (7.17)$$

$$\mu_{y0} = \frac{1}{N} \sum_{i=1}^N y_{i0} - \frac{T\rho\sigma_{y0}^2}{N(T\rho^2\sigma_{y0}^2 + \sigma_u^2 + Ta)} \sum_{i=1}^N (\bar{y}_i - \beta \bar{y}_{i,-1} - \gamma' \bar{x}_i). \quad (7.18)$$

Thus, we may again use a sequential iterative procedure to solve for the MLE. One way to begin the iteration is to use $\hat{\mu}_{y0} = (1/N) \sum_{i=1}^N y_{i0}$ and some arbitrary value of $a/(\sigma_u^2 + Ta)$. When a convergent solution is found we may then solve for σ^2 from a , ρ^2 and σ_{y0}^2 .

For Case III the log likelihood function is

$$\begin{aligned} \log L = & -\frac{NT}{2} \log 2\pi - \frac{NT}{2} \log \sigma_u^2 \\ & - \frac{1}{2\sigma_u^2} \sum_{i=1}^N \sum_{t=1}^T [(y_{it} - y_{i0} + w_{i0}) - \beta(y_{i,t-1} - y_{i0} + w_{i0}) - \gamma'x_{it}]^2 \\ & - \frac{N}{2} \log 2\pi - \frac{N}{2} \log \sigma_\eta^2 - \frac{1}{2\sigma_\eta^2} \sum_{i=1}^N (y_{i0} - w_{i0})^2 \end{aligned} \quad (7.19)$$

For given y_{it} 's (7.19) is a function of w_{i0} 's, β , γ , σ_u^2 and σ_η^2 . If we let $w_{i0} = y_{i0}$, the last term of (7.19) equals zero. As $\sigma_\eta^2 \rightarrow 0$, the likelihood function approaches infinity. The likelihood function does not have a maximum and, hence, the MLE does not exist.

However, we may take partial derivatives of (7.19) and set them equal to zero. Using an argument similar to that of Anderson and Hsiao (1981a) we can show that the solutions for β and γ converge to the CV when $T \rightarrow \infty$, hence they are consistent.

A similar sequential algorithm can be devised to solve for the first-order conditions. Since when T is fixed they are inconsistent and when T tends to infinity they converge to the CV, it seems unfruitful to elaborate on it.

For case IVa the likelihood function is

$$\begin{aligned} \prod_{i=1}^N f_i(y_{i0}, \dots, y_{iT}) = & \prod_{i=1}^N \left(\frac{1}{\sqrt{2\pi}\sigma^2} \right)^{(T+1)/2} |\Omega|^{-\frac{1}{2}} \\ & \times \exp \left\{ -\frac{1}{2\sigma^2} [y_{i0} - \mu_w, y_{i1} - \beta y_{i0} - \gamma'x_{i1}, \right. \\ & \dots, y_{iT} - \beta y_{i,T-1} - \gamma'x_{iT}] \Omega^{-1} \\ & \times (y_{i0} - \mu_w, y_{i1} - \beta y_{i0} - \gamma'x_{i1}, \\ & \dots, y_{iT} - \beta y_{i,T-1} - \gamma'x_{iT})' \left. \right\}. \end{aligned} \quad (7.20)$$

As T goes to infinity and N is fixed, the logarithm of (7.20) will be dominated by terms of (7.7), hence the MLE of β and γ are consistent and the MLE of

λ , σ^2 , μ_w are inconsistent. When T is fixed and N tends to infinity, the MLE is consistent because we are maximizing the likelihood of N independently normally distributed $(T+1)$ component random vectors.

Taking partial derivatives of the logarithm of (7.20) with respect to γ , σ^2 , and μ_w and setting them equal to zero, we have

$$\begin{aligned} \gamma = & \left\{ \sum_{i=1}^N \left[\sum_{t=1}^T \mathbf{x}_{it} \mathbf{x}'_{it} \right. \right. \\ & \left. - \frac{(1-\beta)T^2}{2+(\lambda+T-1)(1-\beta)} \left(\frac{1}{N} \frac{1+\beta}{(\lambda+T)(1-\beta)} \bar{\mathbf{x}}_i \sum_{j=1}^N \bar{\mathbf{x}}'_j - \bar{\mathbf{x}}_i \bar{\mathbf{x}}'_i \right) \right] \right\}^{-1} \\ & \times \sum_{i=1}^N \left[\sum_{t=1}^T \mathbf{x}_{it} (y_{it} - \beta y_{i,t-1}) - \frac{(1-\beta)T^2}{2+(\lambda+T-1)(1-\beta)} \bar{\mathbf{x}}_i (\bar{y}_i - \beta \bar{y}_{i-1}) \right. \\ & \left. - \frac{(1-\beta^2)T}{2+(\lambda+T-1)(1-\beta)} \bar{\mathbf{x}}_i \right. \\ & \left. \times \left(y_{i0} - \frac{1}{N} \sum_{j=1}^N \left\{ y_{j0} - \frac{1}{(\lambda+T)(1-\beta)} (\bar{y}_j - \beta \bar{y}_{j-1}) \right\} \right) \right], \quad (7.21) \end{aligned}$$

$$\mu_w = \frac{1}{N} \sum_{i=1}^N \left[y_{i0} - \frac{T}{(\lambda+T)(1-\beta)} (\bar{y}_i - \bar{y}_{i-1} \beta - \gamma' \bar{\mathbf{x}}_i) \right], \quad (7.22)$$

$$\begin{aligned} \sigma^2 = & \frac{1}{N(T+1)\lambda} \left\{ \sum_{i=1}^N [(1-\beta^2)(y_{i0} - \mu_w)^2 + \sum_{t=1}^T (y_{it} - \beta y_{i,t-1} - \gamma' \mathbf{x}_{it})^2] \right. \\ & - \frac{(1-\beta)}{[2+(\lambda+T-1)(1-\beta)]} \sum_{i=1}^N [(1+\beta)(y_{i0} - \mu_w) \\ & \left. + \sum_{t=1}^T (y_{it} - \beta y_{i,t-1} - \gamma' \mathbf{x}_{it})]^2 \right\}, \quad (7.23) \end{aligned}$$

Substituting (7.23) into (7.20), we have the concentrated log likelihood function in terms of β and λ of the form (4.26). Therefore using the CV of γ and $\hat{\mu}_w = (1/N) \sum_{i=1}^N y_{i0}$ as the initial value, we may search for the MLE by finding the values of β and λ which maximize (4.26). Substituting these values back into (7.21) and (7.22) we can repeat the process until the solution converges.

For Case IVb the likelihood function is of the form

$$\begin{aligned} \prod_{i=1}^N f_i(y_{i0}, \dots, y_{iT}) &= (2\pi)^{-N(T+1)/2} (\sigma^2)^{-N(T+1)/2} |\Phi|^{-N/2} \\ &\times \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^N (y_{i0} - \mu_w, y_{i1} - \beta y_{i0} - \gamma' x_{i1}, \dots, y_{iT} - \beta y_{i,T-1} - \gamma' x_{iT}) \Phi^{-1} \right. \\ &\left. \times (y_{i0} - \mu_w, \dots, y_{iT} - \beta y_{i,T-1} - \gamma' x_{iT})' \right\}, \end{aligned} \quad (7.24)$$

where by letting $\sigma_{w0}^2 = \lambda \sigma^2 / \tau$, we can see that Φ is different from Ω only by the upper left element in which the term $1/(1-\beta^2)$ is replaced by $1/\tau$. Therefore, the MLE will have the same properties as in Case IVa. However a similar sequential computational algorithm does not appear to exist. A Newton-Raphson type iterative procedure will have to be used to obtain the MLE.

In Case IVc where w_{i0} is random with mean θ_{i0} and variance $\sigma_u^2/(1-\beta^2)$, the density function of $y_{i0}, \dots, y_{iT}$ is

$$\begin{aligned} f_i(y_{i0}, \dots, y_{iT}) &= (2\pi)^{-(T+1)/2} (\sigma^2)^{-(T+1)/2} |\Omega|^{-\frac{1}{2}} \\ &\times \exp \left\{ -\frac{1}{2\sigma^2} (y_{i0} - \theta_{i0}, y_{i1} - \beta y_{i0} - \gamma' x_{i1}, \dots, y_{iT} - \beta y_{i,T-1} - \gamma' x_{iT})' \Omega^{-1} \right. \\ &\left. \times (y_{i0} - \theta_{i0}, \dots, y_{iT} - \beta y_{i,T-1} - \gamma' x_{i,T})' \right\}. \end{aligned} \quad (7.25)$$

We note that Case III and Case IVc are similar in the sense that in Case III y_{i0} is normally distributed with mean w_{i0} and variance σ_η^2 ; in Case IVc y_{i0} is normally distributed with mean θ_{i0} and variance $\sigma_u^2/(1-\beta^2) + \sigma_\eta^2$. However, in Case III the MLE does not exist; yet in Case IVc the MLE does exist. This can be seen by using a similar argument showing the non-existence of the MLE in Case III. Write $f_i(y_{i0}, \dots, y_{iT}) = f_i(y_{i1}, \dots, y_{iT} | y_{i0}) f_i(y_{i0})$ and let $\hat{\theta}_{i0} = y_{i0}$, the exponential term of the second density function becomes 1. If we let $[\sigma_u^2/(1-\beta^2) + \sigma_\eta^2]$ approach zero, it means both σ_u^2 and σ_η^2 approach zero. Although $[\sigma_u^2/(1-\beta^2) + \sigma_\eta^2]^{-\frac{1}{2}}$ will approach infinity, the exponential term in the first function approaches zero faster, therefore the likelihood function is bounded, and the MLE exists.

The partial derivative of the logarithm of (7.25) with respect to θ_{i0} is

$$\begin{aligned} \frac{\partial \log L}{\partial \theta_{i0}} &= \frac{(1-\beta^2)}{\lambda \sigma^2} (y_{i0} - \theta_{i0}) \\ &\quad - \frac{1}{\sigma^2} \left(\lambda + T + \frac{1+\beta}{1-\beta} \right)^{-1} \\ &\quad \times \left[(1+\beta)(y_{i0} - \theta_{i0}) + \sum_{t=1}^T (y_{it} - \beta y_{i,t-1} - \gamma' x_{it}) \right], \end{aligned} \quad i=1, \dots, N. \quad (7.26)$$

It is obvious from (7.26) that the MLE of θ_{i0} is inconsistent as T or N or both T and N tend to infinity. Substituting the solution of (7.26) into (7.25) and maximizing it with respect to β , γ , σ^2 and λ , we can show that when T is fixed and N tends to infinity, they are inconsistent.

When N is fixed and T tends to infinity, (7.25) is dominated by terms of (7.7); therefore the MLE of β and γ converge to the CV, thus are consistent. But the MLE of λ and σ^2 are again inconsistent.

In the situation where w_{i0} is random with mean θ_{i0} and variance σ_{w0}^2 (Case IVd), the density function is similar to that of Case III. Therefore, the MLE does not exist. If we confine the MLE to the interior solution, then again when T is fixed and N tends to infinity, the MLE is inconsistent. When N is fixed and T tends to infinity, the MLE of λ , σ^2 and σ_{w0}^2 are inconsistent, but the MLE of β and γ converge to the CV, which is consistent.

The MLE for Cases IVc and IVd are complicated. They are consistent for β and γ only when T tends to infinity. Since in these cases they converge to the CV, it does not appear fruitful to use complicated algorithms to solve for the MLE.

We may summarize the consistency property of the MLE in various situations in table 3.

As one can see from table 3, when $T \rightarrow \infty$ the MLE of β and γ are consistent irrespective of the initial conditions. However, a typical panel usually involves a large number of individuals, but only over a short period of time. As has been shown, different assumptions about the initial conditions call for different methods to obtain the MLE and have different asymptotic properties. Mistaking one case for the other in general will not lead to asymptotically equivalent formulae. Consequently, the misused estimator will be inconsistent no matter how large N is. An investigator, therefore, should be very careful in choosing the appropriate initial condition so that the chosen model would correspond to the true situation.

When T goes to infinity, the MLE of β and γ converge to the CV. When T is fixed, the CV is inconsistent no matter how large N is. The MLE is

Table 3
Consistency properties of the MLE's for state dependence models.^a

		N fixed $T \rightarrow \infty$	T fixed $N \rightarrow \infty$
Case I y_{i0} fixed	β, γ λ, σ^2	Consistent Inconsistent	Consistent Consistent
Case II ^b y_{i0} random	IIa β, γ $\mu_{y0}, \lambda, \sigma^2, \sigma_{y0}^2$	Consistent Inconsistent	Consistent Consistent
	IIb β, γ $\mu_{y0}, \lambda, \sigma^2, \sigma_{y0}^2, \rho$	Consistent Inconsistent	Consistent Consistent
Case III w_{i0} fixed	β, γ $w_{i0}, \sigma_u^2, \sigma_\eta^2$	Consistent Inconsistent	Inconsistent Inconsistent
Case IV ^c w_{i0} random	IVa β, γ $\mu_w, \sigma_u^2, \sigma_\eta^2$	Consistent Inconsistent	Consistent Consistent
	IVb β, γ $\sigma_{w0}^2, \sigma_u^2, \sigma_\eta^2, \mu_w$	Consistent Inconsistent	Consistent Consistent
	IVc β, γ $\theta_{i0}, \sigma_u^2, \sigma_\eta^2$	Consistent Inconsistent	Inconsistent Inconsistent
	IVd β, γ $\theta_{i0}, \sigma_u^2, \sigma_\eta^2, \sigma_{w0}^2$	Consistent Inconsistent	Inconsistent Inconsistent

^aIf the MLE does not exist, we replace it by the interior solution.

^bIIa: y_{i0} independent of η_i ; IIb: y_{i0} correlated with η_i .

^cIVa: mean μ_w and variance $\sigma_u^2/(1-\beta^2)$; IVb: mean μ_w and variance σ_{w0}^2 ; IVc: mean θ_{i0} and variance $\sigma_u^2/(1-\beta^2)$; IVd: mean θ_{i0} and variance σ_{w0}^2 .

complicated and depends on the initial condition. Sometimes we have little information to rely upon in making a correct choice of the initial condition. A simple consistent estimator which is independent of the initial condition will be appealing in its own right, and in addition can be used as the initial value to start the iterative process to obtain the MLE. Balestra and Nerlove (1966) suggested an instrumental variable method. Here we suggest an alternative one.

Such a simple consistent estimator can be found by noting that if we take the first difference of (5.6), we have

$$y_{it} - y_{i,t-1} = \beta(y_{i,t-1} - y_{i,t-2}) + \gamma'(x_{it} - x_{i,t-1}) + u_{it} - u_{i,t-1}. \quad (7.27)$$

Therefore, we may use either $y_{i,t-2}$ or $(y_{i,t-2} - y_{i,t-3})$ as instruments for $(y_{i,t-1} - y_{i,t-2})$ and estimate β and γ by the instrumental variable method. The instrumental variable estimators $\hat{\beta}_{IV}$ and $\hat{\gamma}_{IV}$ are consistent when N tends to infinity or T tends to infinity, or both, and are independent of the initial condition.

After obtaining consistent estimates of β and γ , we may estimate σ_u^2 and σ^2 by

$$\hat{\sigma}_{u,IV}^2 = \frac{1}{2N(T-1)} \sum_{i=1}^N \sum_{t=2}^T [(y_{it} - y_{i,t-1}) - \hat{\beta}_{IV}(y_{i,t-1} - y_{i,t-2}) - \hat{\gamma}'_{IV}(\mathbf{x}_{it} - \mathbf{x}_{i,t-1})]^2, \quad (7.28)$$

$$\hat{\sigma}_{IV}^2 = \frac{1}{N} \sum_{i=1}^N (\bar{y}_i - \bar{y}_{i,-1} \hat{\beta}_{IV} - \bar{\mathbf{x}}_i' \hat{\gamma}_{IV})^2 - \frac{1}{T} \hat{\sigma}_{u,IV}^2. \quad (7.29)$$

Eq. (7.28) is consistent as either N or T (or both) tend to infinity. Eq. (7.29) is consistent only when N tends to infinity.

8. Models with both time-varying and time-invariant exogenous variables

We may combine the results of previous sections to derive the results for models with time-varying and time-invariant exogenous variables. Since it does not involve any new problems, we shall only sketch the results in this section.

We showed in section 2 that when models contain only time-invariant exogenous variables there is no way to distinguish between the case where the individual process $\{w_{it}\}$ has its level determined by observable characteristics [eqs. (2.3) and (2.4)] and the case where $\{w_{it}\}$ is independent of observable characteristics [eqs. (2.5) and (2.6)]. However, we also discussed in section 5 that if time-varying exogenous variables appear in the model, we may separate them in terms of state dependence and serial correlation. Thus, for models containing both time-varying and time-invariant exogenous variables it seems appropriate to formulate models with serial correlation as

$$w_{it} = \beta w_{i,t-1} + u_{it}, \quad (8.1)$$

$$y_{it} = w_{it} + \delta' z_i + \gamma' x_{it} + \eta_{it}, \quad (8.2)$$

and to formulate state dependence models as

$$w_{it} = \beta w_{i,t-1} + \delta' z_i + \gamma' x_{it} + u_{it}, \quad (8.3)$$

$$y_{it} = w_{it} + \eta_{it}. \quad (8.4)$$

Eqs. (8.1) and (8.2) imply that

$$y_{it} = \beta y_{i,t-1} + \delta' z_i + \gamma' x_{it} - \beta \gamma' x_{i,t-1} + \alpha_i + u_{it}, \quad (8.5)$$

and eqs. (8.3) and (8.4) imply that

$$y_{it} = \beta y_{i,t-1} + \delta' z_i + \gamma' x_{it} + \alpha_i + u_{it}. \quad (8.6)$$

Various formulations and interpretations suggested in section 5 for both types of the models can again be applied here without modification.

Using the same manipulation as that in section 3 we can derive conditions for the identification of (8.5) and (8.6).⁸

Proposition 8.1. A necessary and sufficient condition for the identification of (8.5) and (8.6) is that the matrix

$$\begin{pmatrix} \frac{1}{N} \sum_{i=1}^N z_i z_i' & \frac{1}{N} \sum_{i=1}^N z_i \bar{x}_i' \\ \frac{1}{N} \sum_{i=1}^N \bar{x}_i z_i' & \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T x_{it} x_{it}' \end{pmatrix} \quad (8.7)$$

has rank $(K_1 + K_2)$.

Proposition 8.2. A necessary condition for the identification of (8.5) and (8.6) is that $N \geq K_1$.

Proposition 8.3. The serial correlation coefficient of (8.5) or the state dependence coefficient of (8.6), β , is identifiable.

The property of the MLE for each of the cases remains the same as that considered in sections 6 and 7 when we put the coefficients of the time-invariant exogenous variable, δ , into the same group as λ and σ^2 . The CV cannot estimate δ , but gives a consistent and asymptotically efficient estimate of β and γ for the state dependence model when T tends to infinity, and is inconsistent when T is fixed and N tends to infinity.

Simple consistent estimators of β , γ , and δ can also be derived by combining the methods suggested earlier. For instance, for the serial correlation model [eqs. (8.1) and (8.2)] we may use the following four-step estimation method:

Step 1. Eliminate individual effect by subtracting individual mean from (8.2); we have (6.12).

⁸For simplicity we satisfy the condition for identifying λ and σ^2 by assuming $T \geq 2$. For a detailed condition for various cases, see section 3.

Step 2. Obtain a consistent estimate of β by applying standard procedures to (6.12). For instance, we may either use Durbin's method or (6.13).

Step 3. Estimate γ by regressing $(y_{it} - \bar{y}_i) - \hat{\beta}(y_{i,t-1} - \bar{y}_{i,-1})$ on $(x_{it} - \bar{x}_i) - (x_{i,t-1} - \bar{x}_{i,-1})\hat{\beta}$.

Step 4. Estimate δ by regressing $(\bar{y}_i - \hat{\beta}\bar{y}_{i,-1} - \hat{\gamma}'\bar{x}_i)$ on z_i .

It is easy to see that when T tends to infinity the estimator γ is consistent and efficient, but the estimator δ is inconsistent when N is fixed. When N tends to infinity, estimation of γ and δ by this method will be consistent.

Similarly, we may suggest a two-step instrumental variable method to estimate β , γ and δ for the state dependence model (8.6):

Step 1. Taking the first difference of (8.6), we obtain (7.27). Use either $y_{i,t-2}$ or $(y_{i,t-2} - y_{i,t-3})$ as instruments for $(y_{i,t-1} - y_{i,t-2})$ and estimate β and γ by the instrumental variable method.

Step 2. Substitute the estimated β and γ into

$$\bar{y}_i - \beta\bar{y}_{i,-1} - \gamma'\bar{x}_i = \delta'z_i + \alpha_i + \bar{u}_{it}, \quad i = 1, \dots, N. \quad (8.8)$$

Estimate δ by the ordinary least squares method.

9. Conclusions

In this paper we have explored the use of error components models in formulating and estimating dynamic behavioral relationships. McCurdy (1981) in this volume has proposed a model with an error structure that combines factor schemes with time series processes. The error structure in our model is a one factor scheme with serially uncorrelated transitory component. Thus, our model may be viewed as a special case of his. By dealing with this relatively simple case we avoid the problems of identification of the unknown factor loading matrix. However, if the shock is uncorrelated and the factor leading matrix is identifiable in MaCurdy's model, the qualitative conclusions of our model will carry through.

Acknowledgement

We wish to thank Naoto Kunitomo, Kenji Okumura and James Powell for their help in preparing this paper.

References

- Amemiya, T., 1967, A note on the estimation of Balestra-Nerlove models, Technical report no. 4 (Institute for Mathematical Studies in the Social Sciences, Stanford University, Stanford, CA).
- Amemiya, T., 1973, Regression analysis when the dependent variable is truncated normal, *Econometrica* 41, 977-1016.
- Anderson, T.W., 1971, *The statistical analysis of time series* (Wiley, New York).

- Anderson, T.W. and C. Hsiao, 1981a, Estimation of dynamic models with error components, *Journal of the American Statistical Association* 76, 598–606.
- Anderson, T.W. and C. Hsiao, 1981b, Formulation and estimation of dynamic models using panel data, Technical report no. 342 (Institute for Mathematical Studies in the Social Sciences, Stanford University, Stanford, CA).
- Balestra, P. and M. Nerlove, 1966, Pooling cross-section and time series data in the estimation of a dynamic model: The demand for natural gas, *Econometrica* 34, 585–612.
- Chamberlain, G., 1977a, Education, income, and ability revisited, in: D.J. Aigner and A.S. Goldberger, eds., *Latent variables in socio-economic models* (North-Holland, Amsterdam) 143–162.
- Chamberlain, G., 1977b, An instrumental variable interpretation of identification in variance-components and MIMIC model, in: P. Taubman, ed., *Kinometrics: Determinants of socio-economic success within and between families* (North-Holland, Amsterdam) 235–254.
- Chamberlain, G., 1978, On the use of panel data, Paper presented at the Social Science Research Council conference on life cycle aspects of employment and the labor market (Mt. Kisco, New York).
- Durbin, J., 1960, Estimation of parameters in time-series regression models, *Journal of the Royal Statistical Society B* 22, 139–153.
- Goodrich, R.L. and P.E. Caines, 1979, Linear system identification from nonstationary cross-sectional data, *IEEE Transactions on Automatic Control* AC-24, 403–411.
- Hausman, J.A. and W.E. Taylor, 1980, Panel data and unobservable individual effects, Working paper no. 255 (Department of Economics, Massachusetts Institute of Technology, Cambridge, MA).
- Heckman, J.J., 1978, Simple statistical models for discrete panel data developed and applied to test the hypothesis of true state dependence against the hypothesis of spurious state dependence, *Annales de l'INSEE*, no. 30-31, 227–269.
- Lee, L.F., 1979, Efficient estimation of dynamic error components models with panel data, Discussion paper no. 79-118 (Center for Economic Research, University of Minnesota, Minneapolis, MN).
- MaCurdy, T.E., 1981, The use of time series processes to model the error structure of earnings in a longitudinal data analysis, *Journal of Econometrics*, this issue.
- Maddala, G.S., 1971a, Generalized least squares with an estimated variance-covariance matrix, *Econometrica* 39, 23–33.
- Maddala, G.S., 1971b, The use of variance components models in pooling cross section and time series data, *Econometrica* 39, 341–358.
- Mundlak, Y., 1961, Empirical production function free of management bias, *Journal of Farm Economics* 43, 44–56.
- Mundlak, Y., 1978, On the pooling of time series and cross-section data, *Econometrica* 46, 69–86.
- Nerlove, M., 1971, Further evidence on the estimation of dynamic relations from a time series of cross sections, *Econometrica* 39, 359–382.
- Pagan, A., 1973, Efficient estimation of models with composite disturbance terms, *Journal of Econometrics* 1, 329–340.
- Pagano, M., 1974, Estimation of models of autoregressive signal plus white noise, *Annals of Statistics* 2, 99–108.
- Revankar, N.S., 1980, Analysis of regressions containing serially correlated and serially uncorrelated error components, *International Economic Review* 21, 185–199.
- Richmond, J., 1974, Identifiability in linear models, *Econometrica* 42, 731–736.
- Rothenberg, T.J., 1971, Identification in parametric models, *Econometrica* 39, 577–592.
- Swamy, P.A.V.B. and S.S. Arora, 1972, The exact finite sample properties of the estimators of coefficients in the error components regression models, *Econometrica* 40, 261–276.
- Taylor, W.E., 1980, Small sample considerations in estimation from panel data, *Journal of Econometrics* 13, 203–223.
- Trognon, A., 1978, Miscellaneous asymptotic properties of ordinary least squares and maximum likelihood estimators in dynamic error components models, *Annales de l'INSEE*, no. 30-31, 632–657.
- Wallace, T.D. and A. Hussain, 1969, The use of error components models in combining cross-section with time series data, *Econometrica* 37, 55–72.